

Review Form

Paper ID: MA-237

Paper Title: Automated Vision-Based Bolt Handling for Industrial Applications Using A Manipulator

Evaluation to the author(s) (Please mark X in the respective boxes)					
Criteria	Evaluation				
	Exceptional	Very Good	Good	Fair	Poor
1. The relevance of the research problem for the conference topic.		X			
2. Introduction (research aims and contribution, relevant literature, etc.)			X		
3. Conceptual quality (framework, theory, hypotheses, etc.)				X	
4. Methodological quality (design, sample, measurement, method, data, figures, etc.)					X
5. Results (novelty, interpretation, discussion flow, etc.)				X	
6. Discussion (quality and novelty of conclusions and suggestions, etc.)				X	
7. Readability(English expressions, grammar, etc.)		X			

Overall Evaluation (Please mark/highlight the respective box)				
☐	☐	☒	☐	☐
acceptable for publication without changes	acceptable for publications after minor revisions	Review again after major revisions	not acceptable for publication, but author(s) should be encouraged to resubmit	reject

Based on the six evaluations standard, please give your enough comments about the paper's content including method, modeling, experiment, figures, tables, data etc. if there are.
(Please write more than 100 words.)

Strengths of the manuscript:

The use of template matching with Chamfer distance represents a classic yet effective approach, particularly in controlled environments where objects have distinct shapes. Key strengths include simplicity and effectiveness for the specific applications addressed in this work. The following comments are provided to enhance the manuscript further.

Minor concerns and improvement suggestions:

- Methodology Enhancement:** It might be beneficial to employ computer vision techniques

to enhance images for object classification, and exploring deep learning methods could improve accuracy, even with small datasets.

2. **Template and Reference Consistency:** Please consider reviewing the template for equation referencing. Using '(no.)' instead of 'Eq (no.)' and starting sentences with 'Equation (no.)' may enhance clarity. In addition, it might be helpful to ensure reference formatting is consistent, including the use of 'pp' for page numbers and the correct format for conference papers.
3. **Reference Update:** The author(s) might want to reevaluate references [1] and [2], as they could be outdated.
4. **Clarify Experimental Setup:** Provide a clearer experimental setup, specifying the number of sizes and bolt positioning scenarios in each bin. In addition, include R&R gauge measurements to assess the consistency of your results.
5. **Conduct Error Analysis:** Perform error analysis under varying lighting conditions to identify the most effective setup.
6. **Position Estimation Clarity:** Clarify how the estimated position is computed.
7. **Template Matching Considerations:** Be aware of the limitations of template matching with Chamfer distance. While it can classify specific sizes, it may not suit real-world applications with diverse sizes and types. Consider more flexible approaches for better accuracy.
8. **Summarize Related Works:** Summarize the related works section in a concise table for clarity.
9. **Index Terms Specification:** Ensure index terms consist of 3-5 words for optimal clarity.
10. It is suggested to cite the following references to enhance the background of your paper.
Phan Gia Luan and Nguyen Truong Thinh, "Overconstrained Cable-Driven Parallel Manipulators Statics Analysis Based on Simplified Static Cable Model," International Journal of Mechanical Engineering and Robotics Research, Vol. 11, No. 3, pp. 159-165, March 2022. DOI: 10.18178/ijmerr.11.3.159-165

Nguyen Van Quyen, Bui Thi Hai Linh, Duong Quoc Tuan, Nguyen Trung Thanh, and Nguyen Hong Quang, "Dynamic Stability Control and Calculating Inverse Dynamics of a Single-link Flexible Manipulator," International Journal of Mechanical Engineering and Robotics Research, Vol. 12, No. 4, pp. 200-209, July 2023. DOI: 10.18178/ijmerr.12.4.200-209

Mohammed J. Mohammed, Basma A. Abbas, and Sunddus A. Al-Azawiey, "Flexible Underwater Manipulator Modelling Using Intelligent Method" International Journal of Mechanical Engineering and Robotics Research, Vol. 8, No. 6, pp. 860-866, November 2019. DOI: 10.18178/ijmerr.8.6.860-866